🍎 Apple Stock Price Forecasting

Project P-589 | Advanced Time Series Prediction Using Machine Learning

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Project Objective & Scope

Problem Statement

- Stock market prices change rapidly due to various factors. This project uses Machine Learning to predict Apple's stock price for next 30 days using indicators like NASDAQ, S&P 500, interest rate, inflation, and sentiment. It helps investors make data-based decisions.

Objective

- To forecast Apple's stock price using ML.
- To analyze factors like inflation and market trends.
- To create a user-friendly web app for real-time prediction.
- To deploy the model for public access.

Success Goal

- To Deploy a ML Model with high accuracy in forecasting apples stock price and provide a user-friendly web app for real time prediction and data driven decision making

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System Architecture

- Dataset → Preprocessing → Model Training (XG-Boost) → Model Saving (.pkl)

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- streamlit web app→ User Input → Model Prediction → Output Visualization

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- This workflow connects ML model with web interface for real-time use

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Dataset Overview

Data Source & Structure

Historical Apple stock data enriched with key market indicators to capture broader economic context.

- Primary Source: Apple historical data from past years
- Key Features:

- stock_price
- Nasdaq index
- sp500_index
- Inflation rate
- unemployment rate
- interest rate
- market sentiment
- Data collected between 4 AM and 8 PM, excluding weekends.

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Data preprocessing

Steps performed before training:

- Filled missing values
- Removed duplicates
- Handled outliers
- Scaled features using MinMaxScaler
- Sorted data by date/time
- Clean data helps achieve better model accuracy

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EDA

EDA helped identify trends:

- Apple stock highly correlated with NASDAQ and S&P 500
- Inflation and unemployment affect stock prices
- Market sentiment impacts short-term fluctuations
- Used graphs like line plots and heatmaps for analysis
- Detecting Outliers

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Feature Engineering

- Important features affecting Apple's stock:

- NASDAQ Index
- S&P 500 Index
- Inflation Rate
- Interest Rate
- Unemployment Rate
- Market Sentiment

These features were chosen after EDA and correlation analysis

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Model Selection,Building & Performance

Models Evaluated

1

ARIMA

Uses past stock prices to predict future values based on trends and patterns.

2

SARIMA

Handles both trends and seasonal patterns in time-series stock data.

3

XG BOOST

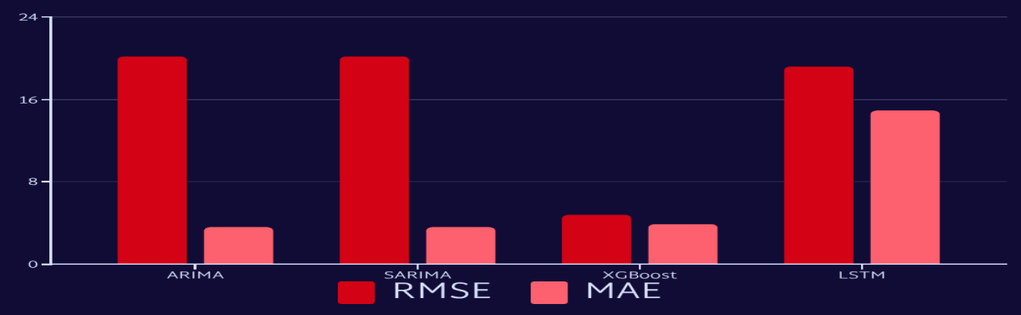
- A powerful machine-learning model that gives highly accurate stock price predictions.

4

LSTM

- A deep-learning model that learns long-term patterns from sequential stock data.

Performance Metrics



XGBoost emerged as the clear winner with significantly lower error metrics across both RMSE and MAE.

1.ARIMA MODEL

RMSE : 20.16

MAE : 3.59

2.SARIMA MODEL

RMSE : 20.14

MAE : 3.59

3.XGBOOST MODEL

RMSE : 4.80

MAE : 3.89

4.LSTM MODEL

RMSE : 19.19

MAE : 14.95

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Best Model: XGBoost Regressor

Superior Accuracy

Lowest prediction error with RMSE of 4.80 and MAE of 3.89

Computational Speed

Rapid training and inference for real-time forecasting applications

Robust Performance

Handles non-linear patterns and feature interactions effectively

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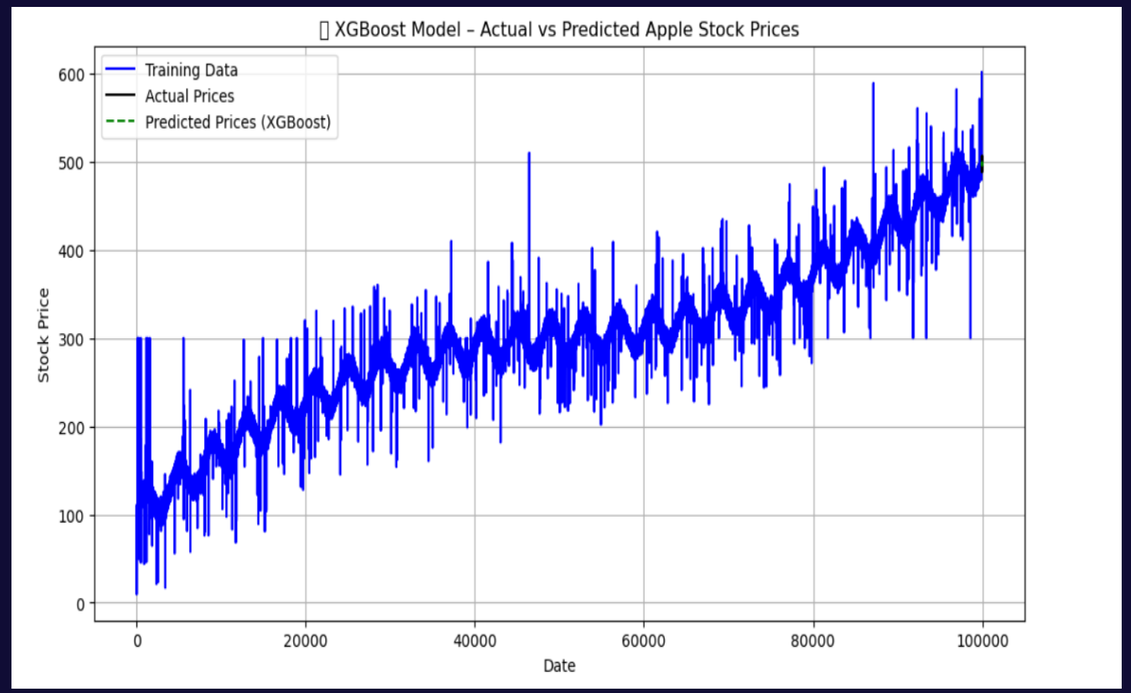
Model Predictions & Forecast

The model demonstrates exceptional alignment between actual and predicted values, validating its forecasting capability on historical test data.

Actual vs Predicted Performance



The forecast projects a moderate upward trend over the next 30 days with expected volatility.

Next 30-Day Forecast

	Day	Predicted Price
0	1	497.849
1	2	497.963
2	3	497.963
3	4	497.849
4	5	497.849
5	6	497.963
6	7	497.118
7	8	497.118
8	9	497.118
9	10	497.118
10	11	497.118
11	12	497.118
12	13	497.118
13	14	497.118
14	15	497.118
15	16	497.118
16	17	497.118
17	18	497.118
18	19	497.118
19	20	497.118
20	21	497.118
21	22	497.118
22	23	497.118
23	24	497.118
24	25	497.118
25	26	497.118
26	27	497.118
27	28	497.118
28	29	497.118
29	30	497.118

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Deployment using Streamlit

Streamlit Deployment

- Interactive web application enabling real-time forecasting with user-friendly interface
- Deployment allows real-time predictions using Streamlit .
- Users can access the model via a browser, enter market details, and get instant price predictions with visuals.

Key Features

Upload custom data, visualize predictions, download forecasts.

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Results after deployment

Apple Stock Price Forecasting App

Upload your dataset (or use the default one) to explore Apple stock trends, run EDA, and forecast prices!

Upload your Excel file

Drag and drop file here
Limit 200MB per file • XLSX

Browse files

Using default dataset: Apples_stock.xlsx

Data Preview

	timestamp	stock_price	nasdaq_index	sp500_index	inflation_rate	unemployment_rate	interest_rate	market_sentiment
0	2010-01-01 00:00:00	102.4836	7984.161	2994.2409	4.5852	6.8374	1.1758	0.0951
1	2010-01-01 01:00:00	99.3472	7995.7062	2995.8318	2.629	5.8514	1.9186	-0.242
2	2010-01-01 02:00:00	103.3156	8000.8877	2998.1196	2.6161	6.2351	2.4367	-0.8336
3	2010-01-01 03:00:00	107.7308	7995.764	3004.5327	3.115	6.3973	2.3691	0.0598
4	2010-01-01 04:00:00	98.9835	8002.4489	2997.1544	4.2703	4.9427	1.93	-0.2231

Dataset Shape: (100000, 8)

GitHub - kharvishrutha/apple | Apple Stock Forecasting - Stre

https://apples-stock-forecast.streamlit.app

Fork

Cleaned and Filtered Data

	timestamp	stock_price	nasdaq_index	sp500_index	inflation_rate	unemployment_rate	interest_rate	market_sentiment
4	2010-01-01 04:00:00	98.9835	8002.4489	2997.1544	4.2703	4.9427	1.93	-0.2231
5	2010-01-01 05:00:00	99.0221	8011.8328	3004.5108	2.3213	4.0956	1.6648	-0.8281
6	2010-01-01 06:00:00	108.1274	8028.1578	3010.7355	2.6261	5.4472	2.9955	-0.8138
7	2010-01-01 07:00:00	104.1071	8027.7223	3015.5049	4.0567	3.3779	3.2763	0.0028
8	2010-01-01 08:00:00	97.9611	8029.5418	3020.1874	2.0998	4.8158	1.7612	-0.6346

Show EDA (Exploratory Data Analysis)

Show Outlier Removal

Run XGBoost Model

Model Performance

RMSE: 4.80

MAE: 3.89

Next 30-Day Forecast

	Day	Predicted Price
0	1	497.8492
1	2	497.9628
2	3	497.9628
3	4	497.8492
4	5	497.8492
5	6	497.9628
6	7	497.1176
7	8	497.1176
8	9	497.1176
9	10	497.1176

✅ Key Achievements

- XGBoost delivered superior accuracy with RMSE of 4.80
- Successfully deployed interactive forecasting platform
- Validated model performance on real-world data

→ Future Enhancements

- Build interactive user interface
- Explore Prophet and advanced LSTM architectures
- Implement automated daily forecast updates

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Thank You

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